Academic year 2023-2024 (Odd semester 2023)

DEPARTMENT OF MATHEMATICS				
Date	**27 December 2023**	Time	**10:00 AM to 11:30 AM**	
CIE II	**Test II**	Maximum Marks	**50**	
Course Title	**FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS**		Course Code	**MAT211BT**
Semester	**I**	Programs	**AS, CH, IM, ME, SPARK**	

Q. No.	Questions	M	BT	CO
1a.	Show that the radius of curvature of the curve $r^n = a^n cos n\theta$ is $\frac{a^n r^{1-n}}{(n+1)}$ for $n \neq -1$.	5	2	1
1b.	Find the centre of curvature of the parabola $y^2 = 4ax$ at the point $(a, 2a)$.	5	2	2
2a.	Using Maclaurin's series expand $e^{sin x}$ up to the terms containing 4th degree.	5	3	2
2b.	If $u = xlog(xy)$ where $x^3 + y^3 + 3xy = 1$ then find $\frac{du}{dx}$.	5	1	1
3a.	If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, then prove that $\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$	5	3	3
3b.	Find $\frac{du}{dt}$ when $u = sin\left(\frac{x}{y}\right)$, $x = e^t$, $y = t^2$.	5	3	3
4.	A rectangular box, which is open at the top, has a capacity of 256 cubic feet. Determine the dimensions of the box such that the least material is required for the construction of the box. Use Lagranges's method of multipliers to obtain the solution.	10	4	3
5.	If $x = \sqrt{vw}, y = \sqrt{wu}, z = \sqrt{uv}$ and $u = r\ sin\theta\ cos\phi, v = r\ sin\theta\ sin\phi$, $w = r\ cos\theta$. Calculate $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$.	10	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	COS / BT	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	10	10	20	10	05	20	15	10	-	-

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Academic year 2023-2024 (Odd semester 2023)

DEPARTMENT OF MATHEMATICS				
Date	20 November 2023	Time	10:00 AM to 11:30 AM	
CIE I	Test I	Maximum Marks	50	
Course Title	FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS	Course Code	MAT211BT	
Semester	I	Programs	AS, CH, IM, ME	

Q. No.	Questions	M	BT	CO
1a.	Reduce the following matrix to an echelon form and hence also, find its rank. $X = \begin{bmatrix} 1 & 3 & 2 & 4 & 1 \\ 0 & 0 & 2 & 2 & 0 \\ 2 & 6 & 2 & 6 & 2 \\ 3 & 9 & 1 & 10 & 6 \end{bmatrix}$	4	2	1
1b.	Test the following equations for consistency and solve if consistent. $x_1 + x_2 - x_3 = 0,\ 2x_1 - x_2 + x_3 = 3,\ 4x_1 + 2x_2 - 2x_3 = 2.$	6	2	2
2a.	Determine the value of b such that the system of equations $2x + y + 2z = 0$, $x + y + 3z = 0$, $4x + 3y + bz = 0$ has (i) trivial solution (ii) non-trivial solution.	4	2	1
2b.	Solve: $3x + 2y + z = 10$, $2x + 3y + 2z = 14$, $x + 2y + 3z = 14$ by the process of Gauss-Jordan elimination.	6	3	2
3a.	Find the angle of intersection of two polar curves C_1: $r = a\theta$ and C_2: $r = \frac{a}{\theta}$, where $\theta > 0$ is the parameter.	4	1	1
3b.	Employ Rayleigh's power method to find the largest eigenvalue and the corresponding eigen vector of the matrix $A = \begin{pmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{pmatrix}$. Assume $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ as the initial eigen vector. Perform three iterations.	6	3	3
4a.	Compute the following for the cardioid which is represented by $r = a(1 - \cos\theta)$ i) Angle between the radius vector and the tangent at any point (r, θ). ii) Angle in which the tangent to curve makes with the initial line. iii) Slope of the curve at $\theta = \pi/6$.	5	3	3
4b.	A curve that is generated by a point on the circumference of a circle as it rolls along a straight line given by the equations $x = t - \sin(t)$ and $y = 1 - \cos(t)$. Obtain the radius of curvature of the parametric curves at $t = \pi$.	5	4	3
5.	Solve the system of equations: $3x_1 + 9x_2 - 2x_3 = 11,\ 4x_1 + 2x_2 + 13x_3 = 24,\ 4x_1 - 2x_2 + x_3 = -8.$ correct up to 4 significant figures by using Gauss-Seidel method after verifying the diagonal dominance of the system. Perform 5-iterations by starting with the initial solution $x^{(0)} = y^{(0)} = z^{(0)} = 0$.	10	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	COS / BT	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	12	12	16	10	04	24	17	05	-	-

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Academic year 2023-2024 (Odd semester)

DEPARTMENT OF MATHEMATICS				
Date	22nd January 2024	Time	10:00 AM to 11:30 AM	
CIE I	Improvement Test	Maximum Marks	50	
Course Title	FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS		Course Code	MA211TB
Semester	I	Programs	AS, CH, IM, ME	

Q. No.	Questions	M	BTL	CO
1a	If $x = e^{r\cos(\theta)}\cos(r\sin(\theta))$ and $y = e^{r\cos(\theta)}\sin(r\sin(\theta))$, then prove that. $$\frac{\partial^2 x}{\partial \theta^2} + r\frac{\partial x}{\partial r} + r^2\frac{\partial^2 x}{\partial r^2} = 0$$	5	1	1
1b	If $z = \sqrt{x^2+y^2}$ and $x^3+y^3+3axy = 5a^2$ find the value of $\frac{dz}{dx}$, when $x = y = a$.	5	2	1
2	The temperature T at any point (x, y, z) in space is given by $T = 400xyz^2$, find the highest temperature on the surface of the unit sphere $x^2+y^2+z^2 = 1$.	10	4	4
3a	Solve the initial value problem $2x''(t) + 5x'(t) - 3x(t) = 0, x(0) = 1, x'(0) = 4$.	5	2	2
3b	Find the general solution of $y'' - 4y = xe^x$.	5	2	2
4	Using Variation of parameters, solve $\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$	10	3	3
5	Solve $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x^2\sin(\log_e(x))$.	10	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	COS / BT	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	10	10	20	10	5	15	20	10	-	-

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

I Semester B. E. Regular / Supplementary Examinations Feb-2024

(Common to AS, CH, IM, ME)

FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.
4. Use of mathematics Handbook is permitted. Do not write anything on handbook.

PART-A

1	1.1	A is a 3×4 real matrix and $A \times = B$ is an inconsistent system of equations. The highest possible rank of A is _____.	01
	1.2	The rank of 7×7 identity matrix is _____.	01
	1.3	Let $A = \begin{bmatrix} 2 & -1 \\ -4 & 5 \end{bmatrix}$, the eigenvalues of A^T are _____ and _____.	02
	1.4	Angle between radius vector and tangent to the polar curve $r = ae^{b\theta}$ $(b > 0)$ is _____.	02
	1.5	Find the radius of curvature at any point t on the curve $x = at^2$ and $y = 2at$.	02
	1.6	If $x = u\sin(v)$ and $y = u\cos(v)$ then $\frac{\partial(x,y)}{\partial(u,v)} =$ _____.	02
	1.7	If $z = x\sin(y) + y\cos(x)$ then $\frac{\partial^2 z}{\partial x \partial y} =$ _____.	02
	1.8	Evaluate $\int_0^3 \int_0^x x\, dydx$	02
	1.9	If $(\bar{x}, \bar{y})$ is the center of gravity of a two-dimensional plate that occupies the triangle $0 \le x \le 1, 0 \le y \le x$ with density function xy and mass $\frac{1}{8}$, then $\bar{y} =$ _____.	02
	1.10	The Wronskian of the functions $u = e^{-3x}$ and $v = e^{x/5}$ is _____.	02
	1.11	The general solution of $y'' + y' + 0.25y = 0$ is _____.	02

PART-B

2	a	Apply Gauss Jordon method to solve the system of equations: $x + 2y + z = 3;\ 2x + 3y + 3z = 10;\ 3x - y + 2z = 13$	05
	b	Find the values of a and b for which the system $x + y + z = 3;\ x + 2y + 2z = 6;\ x + ay + 3z = b$ have i) no solution ii) unique solution iii) infinitely many solutions.	05
	c	Find the dominant eigenvalue and the corresponding eigenvector of the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by Rayleigh power method taking the initial vector as $[1 \quad 1 \quad 1]^T$. Perform five iterations.	06

3	a	Obtain the angle of intersection of the polar curves $r = \frac{a\theta}{1+\theta}$ and $r = \frac{a}{1+\theta^2}$.	08
	b	Find the radius of curvature of the curve $y^2x = a^2(a - x)$, where a is a constant, at the point $(a, 0)$.	08
		OR	
4	a	Obtain Maclaurin series for the function $f(x) = \log_e(1 + \sin(x))$ upto fourth degree term.	08
	b	Find the circle of curvature of the curve $xy = 12$ at $(3, 4)$.	08
5	a	Let $u = 3x + 2y - z$, $v = x - 2y + z$ and $w = x(x + 2y - z)$. Find Jacobian of u, v, w with respect to x, y, z. Interpret the result.	08
	b	If $u = f(r, s, t)$ and $r = \frac{x}{y}, s = \frac{y}{z}$ and $t = \frac{z}{x}$, then find the value of $xu_x + yu_y + zu_z$.	08
		OR	
6	a	If $u = \log_e(\tan x + \tan y + \tan z)$, then find the value of $\sin(2x)\,u_x + \sin(2y)\,u_y + \sin(2z)\,u_z$.	08
	b	A space probe in the shape of the ellipsoid $4x^2 + y^2 + 4z^2 = 16$ enters Earth's atmosphere and its surface begins to heat. After one hour, the temperature at the point (x, y, z) on the probe's surface is $T(x, y, z) = 8x^2 + 4yz - 16z + 600$. Find the hottest point on the probe's surface.	08
7	a	Evaluate $\iint_R xy^2\,dxdy$ where R is the triangular region bounded by $y = 0, x = y$ and $x + y = 2$.	08
	b	Evaluate $\int_0^2 \int_0^{\sqrt{4-x^2}} y^2\sqrt{x^2 + y^2}\,dydx$ by transforming to polar coordinates.	08
		OR	
8	a	Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\,dzdydx$.	08
	b	Evaluate $\int_0^3 \int_1^{4-y} (x + y)\,dxdy$, by changing the order of integration.	08
9	a	Solve the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x^2e^{3x} - \sin(2x)$.	08
	b	The current in an *LRC* circuit is governed by the differential equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = E(t)$. A circuit in series has an electromotive force given by $E(t) = 300V$, a resistor of 10Ω, an inductor of 0.25H and a capacitor of 0.001F. If the initial current and the initial charge on the capacitor are both zero, determine the charge on the capacitor at any time $t > 0$.	08
		OR	
10	a	Using the method of variation of parameters, solve the differential equation $\frac{d^2y}{dx^2} + y = \sec(x)\tan(x)$.	08
	b	Reduce the differential equation $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = x^3 + \sin(\log_e(x))$, where $x > 0$, to a linear differential equation with constant coefficients and hence solve the same.	08